# Multithreaded algorithms exercises

## A16-P01 Work, span, and processor lower bounds

For W=720, D=24, find lower and greedy upper time bounds at P=8,16. Do these metrics alone prove no benefit above30 processors? For 697 independent tasks followed by a23-node chain, prove exact optimum times at P=30,31.

## A16-P02 Recursive matrix work and dependencies

Eight order-n/2 products run simultaneously, then combine through nested parallel loops. Find work, span, and parallelism. Why cannot8 in 8W(n/2) be absorbed away? Where would eight contributions using += on four common C blocks conflict?

## A16-P03 Worst-case parallel-merge span

Derive child size$\leq$3N/4 from the longer-run median and use binary-search cost to prove $O(\log^2 N)$ span. Give a matching lower-bound family. If the other run always stays empty, must span still be $\Theta(\log^2 N)$?

## A16-P04 Scheduling nine Fibonacci strands

Build every dependency of P-FIB(3) in the three-strand model. Find W, D, parallelism, and a time-step greedy schedule for P=3. Prove its optimality. Derive exact general Wn and Dn for n$\geq$2.

## A16-P05 Lost updates and two repairs

Starting x=0, two tasks implement x=x+1 with load/add/store. Give a legal sequentially consistent interleaving ending at 1 and explain why final sync cannot repair it. Compare dependencies and span of atomic addition versus separate writes followed by reduction for n terms.

## A16-P06 Reduction for rectangular matrices

For positive p, q, r and A of size p$\times$q, B of size q$\times$r, give race-free multiplication with valid bounds even when any dimension is 1. Specify full dot-product recursion and outer ownership, then derive work, span, and space under explicit arithmetic/allocation models.

## A16-P07 Longer-run swaps and tie rules

Merge A=[1a], B=[1b,2c], where a, b, c identify original order. Does swapping the longer run first and using key-only lower_bound preserve stability? Trace pivots and destinations. Repair using(key, original position) and give the general reason.

## A16-P08 Output offsets and aliasing

Apply P-MERGE to T=[1a,4b,6c,9d | 2e,3f,6g,8h] with o=3. Find root m, j, q, child output intervals, and final O[3..11). If output instead aliases T with o=0, which unread input does the first root write destroy?

## A16-P09 General balanced-tree charging

A recursion on N records removes one unique record per nonempty call. Its disjoint children each have size$\leq \rho m$ for fixed$0<\rho<1$; nonrecursive cost is O(log(m+1)). What child-count assumption is needed to include empty-call costs? With it, prove O(N) work by distributing charges. Give a chain counterexample when ρ may approach1 with m.

## A16-P10 One input family must realize worst sort span

Why does summing independently worst merge spans not establish a sort lower bound? Construct the n=16 bit-reversed key order, give the root's sorted runs, and explain recursive similarity to obtain worst span $\Theta(\log^3 n)$.

## A16-P11 Reversing the work/span preference

Correct algorithms A, B have(W, D)=(2048,8),(1536,32). Using only work/span lower bounds and greedy W/P+D upper bounds, find integer-P ranges guaranteeing B faster and A faster. How should the intervening range be described?

## A16-P12 Serial spawning differs from parallel for

Batch n unit iterations into groups of g, with ceil(n/g) groups. Compare work/span of an ordinary loop spawning groups versus balanced PAR-FOR. Which g gives the first method's best asymptotic span? Count the parent's task-issuing chain.

## A16-P13 Matrix multiplication with less temporary storage

After parallel zeroing of C, design ACC(C, A, B) for C←C+AB without an $n \times n$ temporary. Use two synchronized phases per C block, with four different blocks concurrent within a phase. Give full recursion, correctness, work/span, space, and comparison to full eight-way expansion.

## A16-P14 Two meanings of average random-DAG speedup

Equally likely DAGs have respectively W=100, D=10, T10=10 and W=100, D=100, T10=100. Compare E[W/T10] and E[W]/E[T10], and give realizable DAGs. Derive an expected-time bound from the pointwise greedy theorem. Must W and D be independent?

## A16-P15

For a balanced reduction of eight inputs, count additions per level and find W and S. Assume unit-cost additions and free spawning.

## A16-P16

For W=100, S=12 and P=4, find the work/span lower bound and greedy Brent upper bound.

## A16-P17

What problems arise from tasks that are too fine or too coarse? Name two costs outside the ideal model.

## A16-M04 Buffer initialization

Suppose each temporary buffer of length m is serially zero-initialized by one thread before recursion. Prove the sort span is at least $\Omega(n)$, explain why $\Theta(\log^3 n)$ no longer applies, and propose a buffer setup compatible with the ideal model.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

## References and attribution

Cormen, Leiserson, Rivest, Stein, Introduction to Algorithms, third edition, Chapter 27. MIT 6.046J Fall 2004, Lecture 3, matrix multiplication and Strassen. MIT 6.046J Fall 2005, A Minicourse on Dynamic Multithreaded Algorithms, Handout 29.

Original educational synthesis by Lin Cing. Original material is licensed under CC BY 4.0. The referenced textbook is not distributed with these notes.

https://github.com/xinhi6716/computer-science-self-study

Lin Cing
https://github.com/xinhi6716/computer-science-self-study